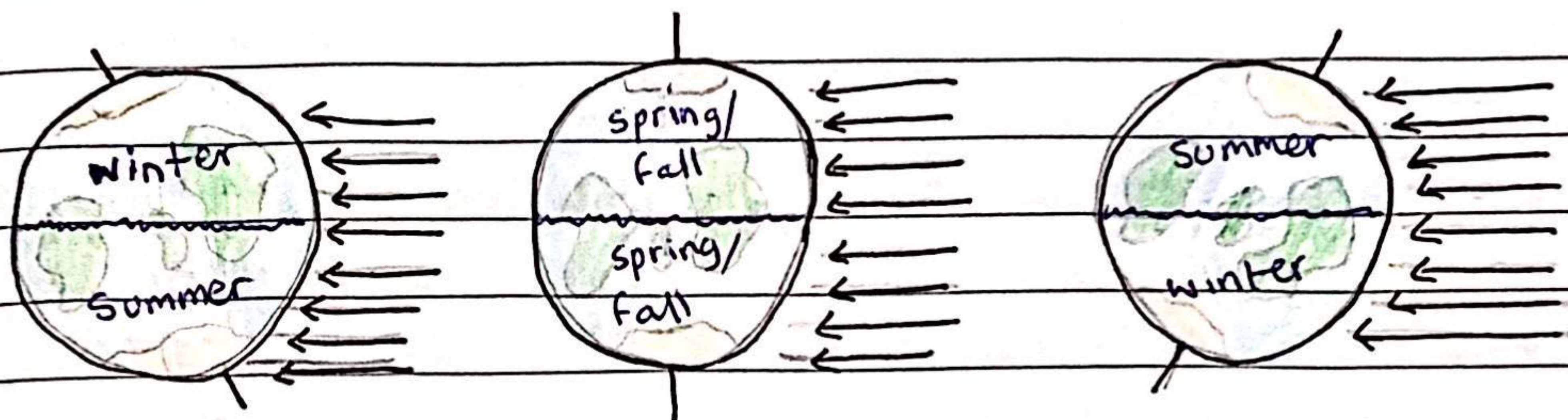


Earth:

a planet that rotates on its tilted axis once in approximately 24 hours.
 ↳ the side facing the sun experiences the daytime while the opposite side experiences nighttime.

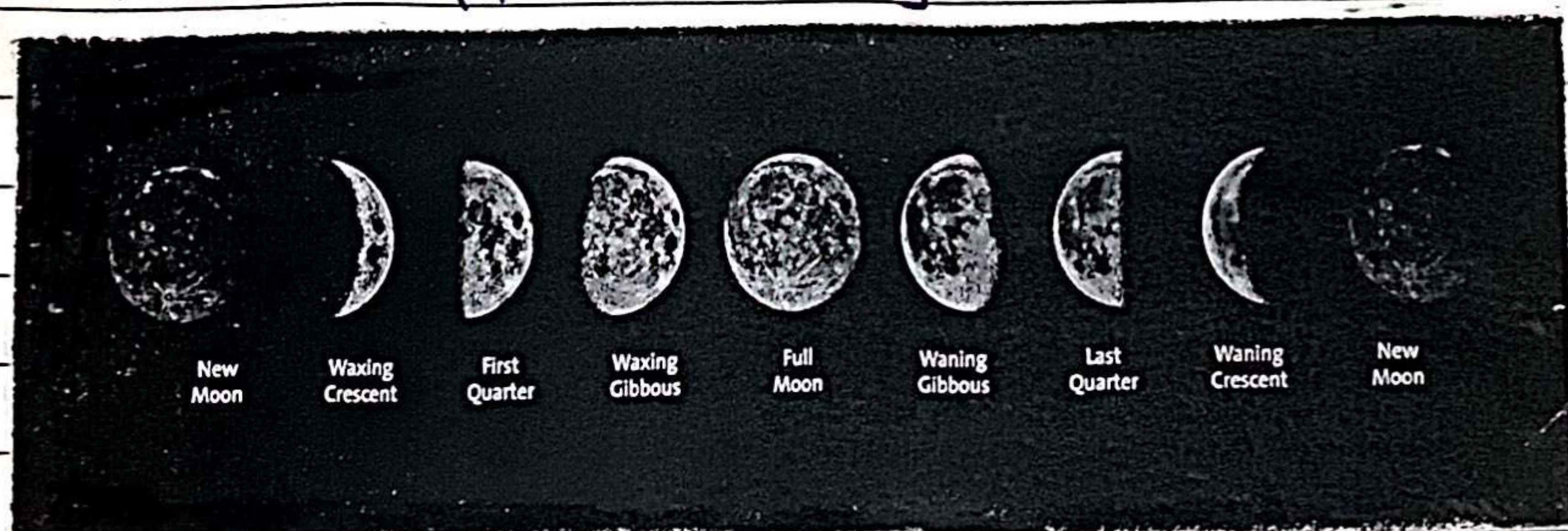
the Earth completes one orbit around the sun once in approximately 365 days



* average orbital speed $\Rightarrow v = \frac{2\pi r}{T}$
 radius $\swarrow$
 orbital period $\nwarrow$

Moon:

the moon completes one orbit around the Earth in approximately one month (28 days)



PHASES OF THE MOON

↳ as the moon orbits the earth, the sun lights up different parts of it, which causes the phases of the moon

Solar system :-

- has one star, the Sun

- ↳ mercury, venus, earth, mars, jupiter, saturn, uranus, neptune

- ↳ dwarf planets such as pluto

- ↳ asteroid belt between mars and jupiter

- ↳ moons that orbit planets

- ↳ smaller bodies such as comets & natural satellites

- * planets, minor planets and comets have elliptical orbits

- ↳ the sun is not at the centre of the elliptical orbit unless the orbit is approximately circular

mercury —————→ neptune

- ↳ orbital distance increases

- ↳ orbital duration increases

- ↳ density decreases

- ↳ surface temp. decreases

- ↳ gravitational f.s depends on the size of the planet

- the 4 planets nearest to the sun are rocky and small & the 4 planets farthest from the sun are gaseous and large.

5 billion years ago...

- a nebula began to collapse in on itself

- ↳ a giant cloud of dust + gases (mostly hydrogen and helium)

atoms collided together - generating heat
↳ hot enough for nuclear fusion ... sun was 'born'
material in nebula not absorbed into the sun, swanned around in a flat disk of dust + gas held in orbit by sun's gravity

↳ Flat disk = accretion disk

each planet began as a microscopic grains of dust in the accretion disk

↳ atoms + molecules stuck together

↳ objects up to a mile in diameter formed
→ attracted together by gravity

the sun's stellar winds blew off most of the gases of the 4 planets nearest to the sun

↳ hence mercury, venus, earth, mars formed
→ small & rocky + metals

the stellar winds could not blow off the gases of the 4 planets farthest from the sun

↳ hence jupiter, saturn, uranus, neptune
→ large & gaseous (small rocky core)

* the strength of the gravitational field depends on the mass of a planet, and it decreases as you go away from the planet.

→ speed of light = 3.0×10^8 m/s

→ to calculate the time = $\text{distance} / \text{speed}$

* the sun contains most of the solar system's mass which means it has the strongest gravitational force thus concluding why all the planets orbit around it.

05/01/23

* the strength of the sun's gravitational field and the orbital speed decreases as the distance from the sun increases

an object in an elliptical orbit travels faster when closer to the sun

↳ conservation of energy

→ when the object comes closer to the sun, it gains E_k , but for that it loses E_p and when it moves away from the sun, it gains E_p , but for that it loses E_k .

* the Sun is a star of medium size, consisting mostly of hydrogen & helium. It radiates most of its energy in the infrared, visible and ultraviolet region

↳ stars are powered by nuclear reactions that release energy. In stable stars the nuclear reactions include the fusion of hydrogen into helium

The Universe :-

galaxies are each made up of billions of stars

↳ the sun is a star in the 'Milky Way'

↳ other stars in the Milky Way are much further away from Earth than the sun

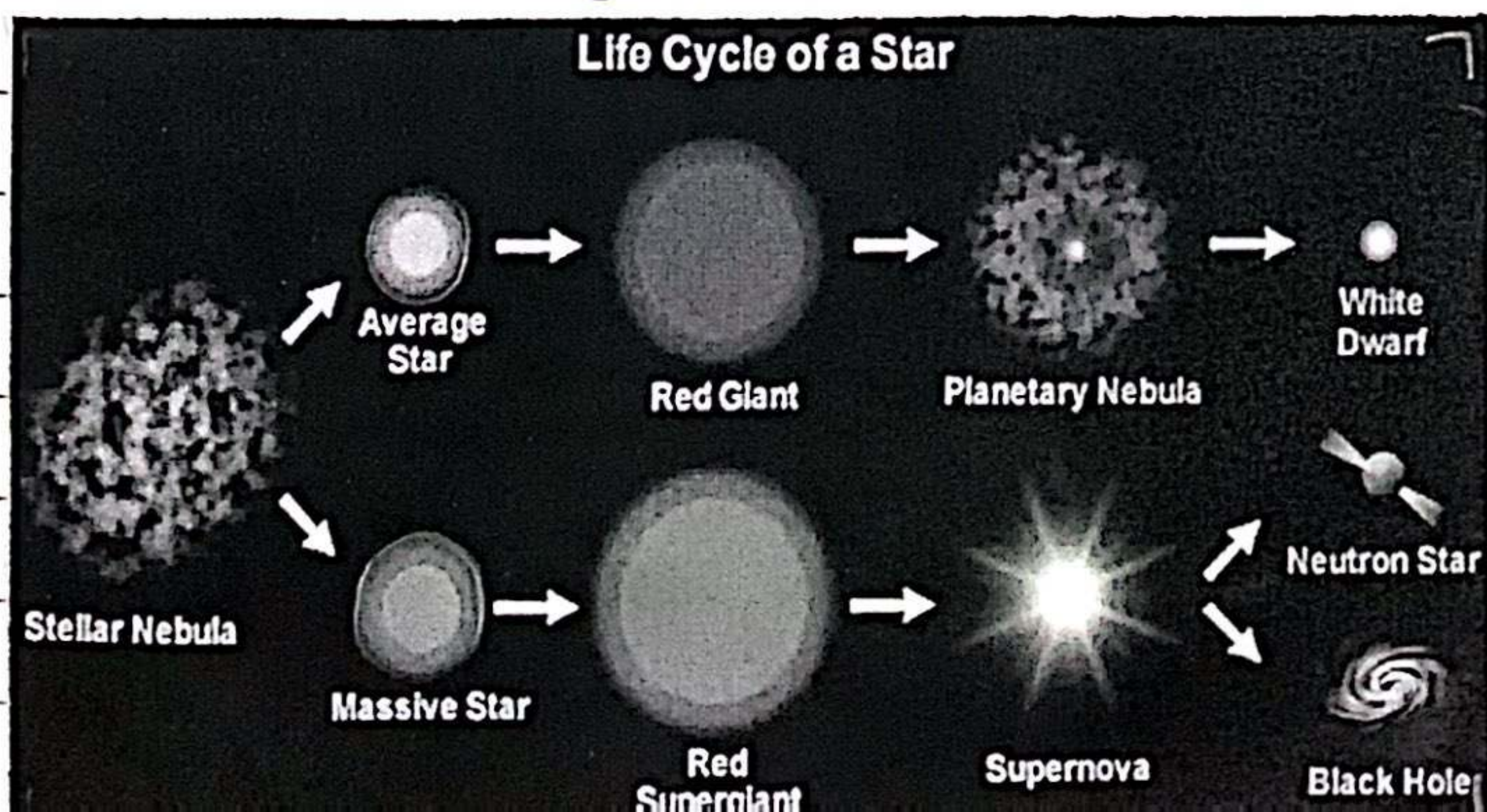
* astronomical distances can be measured in light-years.

↳ one light-year is the distance travelled by light (in a vacuum) in one year

↳ one light-year = 9.5×10^{15} m

Lifecycle of a star:-

- a star is formed from interstellar clouds of gas and dust that contain hydrogen
- ↳ a protostar is a interstellar cloud collapsing and increasing in temperature because of its internal gravitational attraction
- ↳ a protostar becomes a stable star when the inward force of gravitational attraction is equal to an outward force due to the high temperature in the centre of the star
- ↳ a stable star undergoes nuclear fusion where hydrogen turns into helium, which releases heat/light energy that keeps the core hot
- ↳ most stars expand to form red giants and more massive stars form red supergiants when most of the hydrogen converts to helium
- ↳ a red giant forms a planetary nebula
- ↳ a planetary nebula turns into a white dwarf
- ↳ a red supergiant explodes as a supernova, (forming a nebula) leaving behind a neutron star or a black hole
- ↳ the nebula may form new stars with orbiting planets.



* the Milky Way is one of the many billions of galaxies making up the universe
↳ its diameter is approx. 100 000 light-years

Redshift

- an increase in the observed wavelength of electromagnetic radiation emitted from receding stars & galaxies
- light emitted from distant galaxies appears redshifted in comparison with light emitted on Earth
 - ↳ this means that the galaxies are moving away from us
 - ↳ which means that the universe is expanding
 - ↳ this also supports the Big Bang Theory

* [microwave radiation of a specific frequency
① is observed at all points in space around us] and is known as cosmic microwave background radiation (CMBR)

- ① ↳ CMBR was produced shortly after the universe was formed
- ① ↳ it expanded into the microwave region of the electromagnetic spectrum as the universe expanded

- (v)
→ the speed at which a galaxy is moving away from Earth can be found from the change in wavelength of the galaxy's starlight (redshift)
- the distance of a far galaxy (d) can be found by the brightness of a supernova in that galaxy

Hubble constant: (H_0)

- the ratio of the speed at which the galaxy is moving away from Earth to its distance from the Earth

$$* H_0 = \frac{v}{d}$$

→ the current estimate for H_0 is:
 2.2×10^{-18} [per sec]

* if you reverse the equation:

$$\frac{d}{v} = \frac{1}{H_0}$$

→ this represents an estimate for the age of the universe

→ also its evidence that all matter in the universe was present at a single point